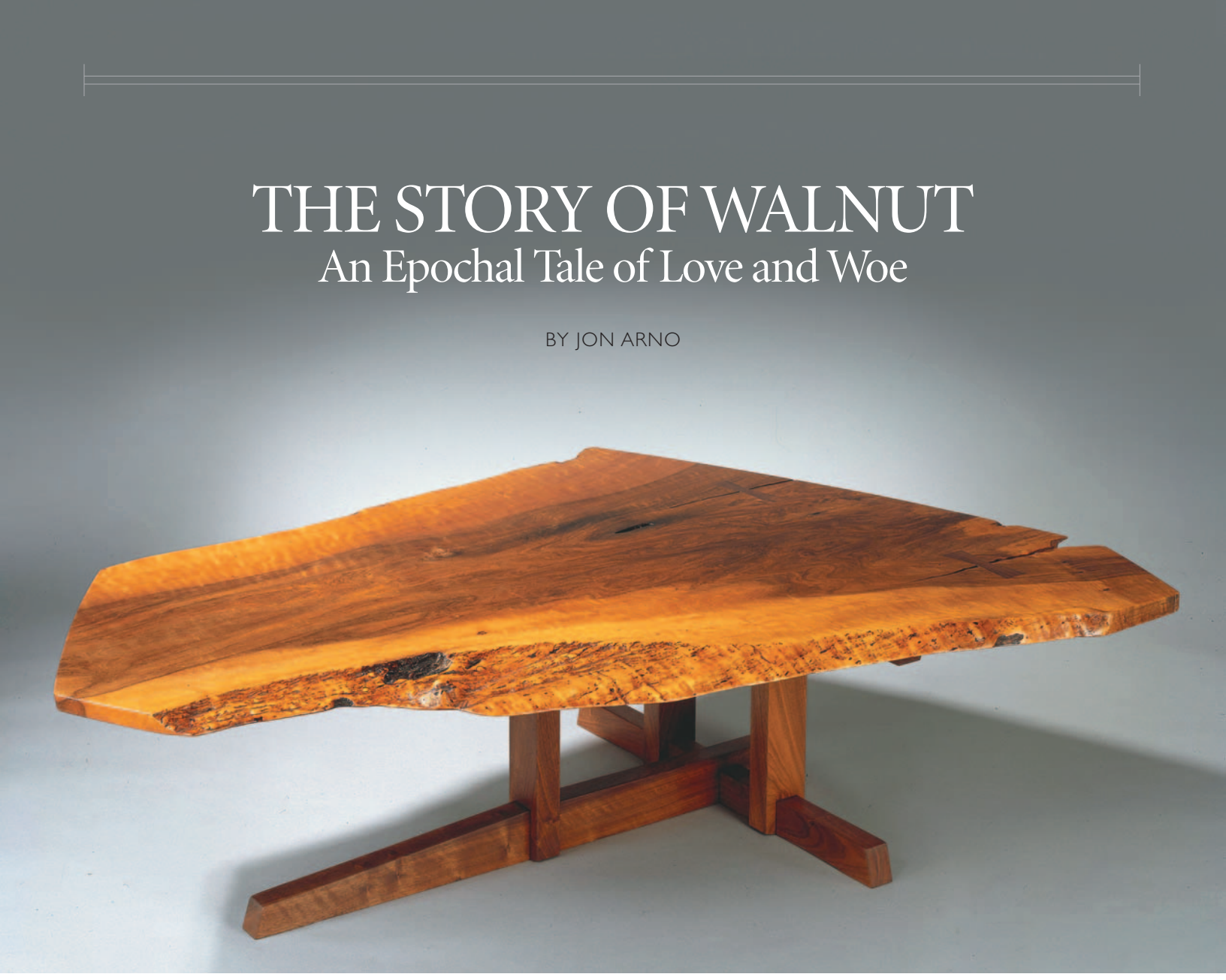
THE STORY OF WALNUT
An Epochal Tale of Love and Woe

BY JON ARNO



No furnituremaker celebrated the beauty of walnut more than George Nakashima; the top of this 1966 coffee table is a slab of English walnut.

To woodworkers around the world, there is something very special about walnut. Curiously though, opinions differ widely as to just exactly what it is that makes it so special. From the American perspective, the dark chocolate brown color and greater strength of our native black walnut sets it apart. As styles come and go, at least when dark wood is in, our walnut reigns supreme. On the other hand, European woodworkers defend the flip side of this argument. Their slightly softer species of walnut, the so-called "English" walnut, has excellent shaping characteristics and possesses a more middle-of-the-road, grayish milk chocolate color. What's more, it's pigmentation is often variegated, blending both light and dark brown marble-like swirls, which gives it the flexibility to be used in contrasting combinations with either light or dark accent woods. No doubt about it, both of these walnuts are magnificent cabinet woods, but if the debate proves anything, it is only the provincial arrogance of our Western Culture, regardless of which side of the Atlantic you're on.

In fact, the world is blessed with far more than just these two species of walnuts, and there isn't a loser in the bunch. The better approach to understanding the walnuts (of which there are at least 15 species around the globe) is to first address what characteristics they have in common. For example, they are all relatively stable woods with low T/R ratios; that is, their tangential and radial shrinkage properties are less divergent than most woods, and therefore they have a lower propensity to distort when exposed to changes in humidity. With only one major exception (butternut), they all enjoy a sort of splendid mediocrity in terms of density: soft enough to be comfortably worked with hand or power tools and yet strong enough for most furniture applications. In fact, with respect to their ability to absorb shock, they tend to be much more endowed than other woods with comparable densities…hence, they make outstanding gunstocks. Although they are mostly moderately coarse-textured woods, they yield a waxy and somewhat lustrous surface when planed or carved, and will hold relatively crisp edges when shaped. Also, they all tend to have at least moderate, or more often very good decay resistance.

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PHOTO BY JAY ODEE

Sam Maloof's rocker (1975) uses walnut carefully selected for uniformity of color in order to emphasize the sculptural lines of the work.



Taimi Barty's sushi trays (2003) are made from orchard walnut at the graft line between California black walnut and English brown walnut.

Wendell Castle's enormous "Library Sculpture" (1965) achieves much of its visual impact from the multiple laminations of walnut in all its colorings, rather than seeking a uniform tone.

However, their durability in exterior applications does vary, and this characteristic appears to correlate closely with their pigmentation. The darker species seem to have the greater resistance…but we'll deal with the natural chemistry of the walnuts a little later.

Setting aside the debatable merits of color for a moment, all of the walnuts share a very attractive appearance based on the similar anatomical structure of their figure. Again, it is a beauty borne of mediocrity; the walnuts are neither sharply ring-porous like ash or oak, nor are they even remotely diffuse-porous like maple or birch. When viewed on the endgrain, all of the walnuts are best described as semi-ring-porous woods. They all display the characteristic band of relatively large pores along the annual ring, like the truly ring-porous woods, but as the growing season progresses, there is no abrupt transition to the denser summer wood tissue. Instead, the pores gradually decrease in diameter and tend to become fewer in number and more widely spaced. What this anatomical structure presents on the flatsawn surface is a figure with a recognizable character, but one that is more mellow than bold. In that inevitable trade-off between the striking figure of ring-porous woods and the more friendly workability (especially shapeability) of the diffuse-porous woods, the walnuts again hold the middle ground. They stand and deliver what is arguably the perfect compromise.

If we accept for the moment that every walnut is a winner, it opens the broader and more complex issue of…just exactly what is a walnut? Offering an answer to this question risks steeping the conversation in more botany than the average woodworker cares to know, but for those with a burning curiosity about the genesis of this world-class group of timbers, it's worth the trip.

THE WALNUT FAMILY TREE

Fossilized pollen dates ancestors to the modern walnut (Juglandaceae) from the late Cretaceous, about 75 million years ago. Botanists quibble about the begats and begottens of these ancient angiosperms, but the walnuts seem to stand apart from the more closely related beeches, chestnuts and oaks in the beech family (Fagaceae), and also from the birches (Betulaceae). Although the walnuts probably deserve their own Order (Juglandales), they obviously share some affinity with this broader group of nut bearers—possibly more so with the hazelnut side of the birch family—but we really don't know their exact evolutionary position. Both the make up of the family and the distribution of its surviving species suggests great antiquity and a now long lost preeminence.

The walnut family contains only six surviving genera (*Alfaroa, Carya, Engelhardtia, Juglans, Platycarya* and *Pterocarya*) and a total of somewhere between 50 and 60 species. From the woodworking perspective, two of these genera—*Alfaroa* and *Platycarya*—can be quickly dismissed, since they aren't significant lumber sources. The other four genera all do play some role in woodworking, even though two of them are virtually unknown to the craft in North America.

The wingnut genus (*Pterocarya*) is native to the Caucasus and eastward in pockets all the way to China. Wingnut typically develops into a multi-trunked tree that isn't useful for lumber production, but the wood is much appreciated by artisans within its native range for making small

items and for carving. The wood is moderately soft, light creamy tan in color, and sometimes displays pinkish highlights. It is unusually fine-textured for a walnut, and more comparable in appearance and working properties to basswood or aspen.

The genus *Engelhardtia* is primarily of tropical distribution, found both in southern Asia, from India southward and eastward into Indonesia and the Philippines, and also in Central America. The woods of the various *Engelhardtias* (of which there are about 17 species) also tend to be moderately fine-textured compared to other members of the walnut family…and, as you might expect, they have correspondingly excellent shaping characteristics. Usually displaying a light grayish tan color, sometimes with greenish highlights, the *Engelhardtias* are a bit denser than wingnut, ranging from about that of yellow poplar up through some of the denser soft maples. In India, where this wood is known as "silapoma," and also throughout Southeast Asia, they are commonly used for utilitarian turnings such as kitchen utensils and bowls, as well as in general construction applications. The two Central American species of E*ngelhardtia* have similar characteristics and the wood usually goes by the common name of "gavilan."

Far and away, the only two genera in the walnut family that you might justifiably call heavy hitters in the international lumber trade are *Carya*, the hickories, and *Juglans*, the walnuts. With the exception of one or at most two species found in eastern and southern China, the remaining dozen or so hickories are exclusively native to eastern North America, from northern Mexico northward into southeastern Canada. The hickories are the unchallenged muscle woods of the walnut family, possessing high density and exceptional elasticity, but they are such an important group of timbers that they deserve a separate article of their own, so we will only mention them here in passing. The key point is, now that we've surveyed all of its shirtsleeve relatives, the stage is set for our primary topic, the genus *Juglans*.

PARADISE LOST

In the grander geological scheme of things, the sporadic distribution of our modern walnuts seems to tell a tale of serious recent decline. Because the walnuts hybridize easily, it is difficult to pin down the exact number, but all together the total probably ranges somewhere between 15–20 legitimate species, scattered throughout the world. There were once substantially more species in the *Juglans* genus, but just how much their numbers have shrunk over the past couple of million years we cannot be entirely sure. However, what we can be absolutely certain of is that for a very long time a very diverse and plentiful population of walnuts could be found almost everywhere in the Northern Hemisphere.

How could the walnuts be so successful for so long and then cave in so quickly? For at least part of the answer, we need only look to the natural attributes of the walnuts themselves. First, in terms of their long success, the walnuts can be intensely competitive because of their unique chemistry. They contain an extractive called juglone that has a number of interesting properties. It is an antiseptic that helps provide decay resistance to the wood, though unfortunately, it is a potentially serious allergen for some woodworkers. It is also the same extractive that polymerizes to give the wood its dark color, but its more important role from the perspective of the living tree is that it is one of the most effective natural herbicides in the Plant Kingdom. Many species from several very dissimilar plant families are extremely intolerant of juglone and are either stunted by it or die out altogether when they attempt to grow within the drip line of walnut trees. Interestingly, this is especially true of the pines, which walnut must have competed with early on in its evolution. So, walnut is what you might call a bully in mixed species forests, and it is hardy in other ways as well. It is rel-

Air-Dried or Kiln-dried

OPINIONS VARY as to whether air-dried or commercially steamed and kiln-dried walnut provides the best cabinet wood. Proponents of air drying often cite the beautiful purple highlights found in air-dried stock as a benefit that is lost when this species is steamed. Also, air-dried walnut seems to retain a waxy feel, giving it better shaping qualities than steamed stock. Conversely, proponents of steamed and kiln-dried walnut favor it because the steaming process makes the stock more uniform in color. Furthermore, kiln drying relaxes the shrinkage stresses the wood experiences when exposed to the seasonal, low humidity extremes most furniture must endure in modern, centrally-heated homes. For commercial furniture manufacturers, this translates into less sorting at the factory in order to color match the material used in each finished piece of furniture…and, eventually, fewer customer complaints resulting from warping and checking when the furniture is exposed to its final in-use environment. Also, the lumber industry and most conservationists favor using the steaming process because it makes more efficient use of the resource. Steaming darkens the sapwood, making more of the log usable as cabinet stock. With most species, the additional processing costs would offset this conservation advantage, but walnut is unique in one very important respect. This species naturally converts sapwood to heartwood very slowly and even fully mature logs often retain a sapwood band that is upwards of 4" wide. To scrap such a large portion of the log is extremely wasteful when the sapwood's only functional disadvantage is its light color.

So, who is right? As is the case with most controversies, there are tradeoffs. In my experience, the superior shaping qualities of air-dried walnut are lost in the kiln-drying process. However, the superior color of air-dried stock—more variegated with beautiful purple highlights—is, at best, a transient advantage. The short wavelength "blue" pigments in walnut that combine with red pigments to create the purple highlights are fugitive and ultimately succumb to the wood's natural patina-forming process. Whether air-dried or steamed, walnut eventually settles into this species' stable, long-term, dark chocolate brown color and there is nothing you can do to permanently stop the transition. While I personally much prefer working with air-dried stock, my conscience tells me conservation is better served the other way.

atively tolerant of seasonal fluctuations in temperature, actually preferring cool climates that experience some periods of freezing weather. Unfortunately, what appears to have made it such a Pleistocene patsy is that it is lethargically slow when it comes to dispersion. It just couldn't deal with the Ice Age.

Unlike species with wind-borne seeds (such as the willows) or those with fruits that are eaten relatively whole (like the cherries) and then spread by migrating birds, walnut just can't expand its range very quickly. It puts a great deal of energy into producing relatively few but very large nuts that tend to thud to the ground directly below the parent tree. While the occasional uncommonly ambitious squirrel might carry a nut a few hundred feet before burying it, walnuts have gotten precious little help from the Animal Kingdom when it comes to expanding their territory…that is, until the arrival of mankind in very recent times.

When the Pleistocene glaciers advanced southward, it was difficult for walnut to retreat, or to regain territory during interglacial periods. This was made doubly difficult in Europe because of the east-west alignment of the mountain ranges all the way from the Pyrenees in northern Spain to the Caucasus in southern Russia. As a result, walnut was trapped and wiped out from all of Europe north of the Alps and west of the Urals. It probably owes its survival in North America to the predominantly north-south orientation of our mountain ranges, and in the extreme eastern Orient, because of the lower elevations of the mountains and simply because the glaciation was less severe there.

FROM PERSIA WITH LOVE

Walnuts have long been an important part of the human diet and they are a very portable, high energy food. We have carried them with us wherever we have gone and both we and they appear to have benefited from the relationship. According to conventional wisdom, at the end of the last glaciation some 10,000 years ago, the one remaining species of walnut, *Juglans regia*, was clinging to survival by the tips of its twigs somewhere between the shores of the Black Sea and the Caspian in northern Persia. Then, on the eve of historical times, it was brought westward by humans. While it is likely that this Persian walnut did not reach Britain until the time of the Roman occupation a mere 2000 years ago, it is far more likely that it was brought into the Mediterranean region well back in the Stone Age. In any event, mankind has clearly aided in the more rapid proliferation of this species, certainly in a westward direction and possibly also toward the east, where it is now found sparingly in northern China.

As for the American walnuts, they too have migrated northward at least a few hundred miles since the end of the last glaciation, but whether this has been accomplished by squirrel power alone or with the help of humanity is a little less obvious. Certainly, walnuts were an important part of the diet of native Americans, and the eastern woodland tribes were fairly mobile cultures. So, this love affair between humans and the walnut was probably global in breadth.

SORTING THE WALNUTS

The woods of our two native eastern species, black walnut and butternut, pretty much represent the extremes of variability within the *Juglans* genus in terms of both density and color. To be careful though, the operative phrase here is "pretty much," since one of the Andean species (*J. neotropica*, which is often seen on the international market and sold as "Peruvian walnut" or "nogal") does occasionally produce wood that is slightly darker in color than our

How Steaming Works

IT WAS LONG BELIEVED that steaming walnut darkened the sapwood primarily by causing some of the heartwood pigments to migrate into the sapwood. This notion of borrowing from Peter to pay Paul seemed plausible since heartwood highlights were lost in the steaming process, and both the heartwood and sapwood became more uniformly brown (less variegated) in color. However, controlled and carefully measured tests conducted in the 1960s revealed the chemistry involved is far more complex than simple pigment migration. It was found that pure walnut sapwood, trimmed of any associated heartwood, could still be coaxed to darken when exposed to steam. In other words, pigments were being formed and not merely moved around.

This unexpected result next led the researchers to investigate which aspects of steam (the heat and/or the moisture) actually caused pigment formation. They found that moisture did play some role in the process: steaming was more effective in freshly-cut green wood with moisture content above the Fiber Saturation Point than it was with stock that had been air-dried prior to steaming. On the other hand, heat seemed to be the primary trigger in accelerating pigment formation. In fact, when the pressure was increased in the steam chamber to allow for temperatures above the boiling point of water, the pigment-forming process appeared to be optimized as the temperature approached 230° F. Whether the moisture simply helps transmit the heat or actually contributes chemically to the pigment formation is still not clear.

It is thought that the dark color of walnut results from an enzyme action on some of the wood's natural phenols, causing them to polymerize into pigments. This process is accelerated by heat, but heat is not indispensable. Nature achieves the same result at ambient temperatures as sapwood slowly converts to heartwood in the living tree. Also, while steaming quickly breaks down the beautiful highlights found in walnut, over time nature destroys them anyway, as the wood takes on its long-term patina. So, while modern technology allows us to tamper with walnut's chemistry and force a speedier transformation of its natural pigment-forming process, the mystery remains as to how it actually works.

Research consulted:

"Color Changes in Black Walnut As a Function of Temperature, Time, and Two Moisture Conditions"; by A. B. Brauner and W. E. Loos, *Forest Products Journal*, vol. 18 #5, May 1968.







The great size and good carving properties of black walnut were used by Jon Brooks to create this seat from 1971.

Walnut shines even as a "secondary wood": Art Carpenter's rolltop desk (1970) mixes walnut with mutenye, a West African hardwood, for a nice visual effect.

In this Wharton Esherick music stand (1951), the color and texture of the horizontal walnut pieces nicely complement the vertical lines of cherry.

black walnut. Nogal's color is quite variable, however, and it differs from black walnut in other important ways, being almost always coarser in texture and slightly lighter in weight. With that one possible exception though, our black walnut (*J. nigra*) is both the darkest and densest of all the commercially important species. On the other hand, without exception, our butternut (*J. cinerea*) is the least dense and lightest in color. Actually, there seems to be three natural groups within the walnut genus; those species producing dark woods of the black walnut type, those which are of the blond, butternut variety, and those of a third, variegated type, of which only the Persian walnut is so far a confirmed member.

The differences between the groups are readily apparent in the appearance of the woods, but there is more to it than mere superficial appearance. Based on the long experience of nut growers, there also seems to be some genetic underpinnings that support this sub-typing. It is known that species within the black walnut group hybridize with each other without much difficulty, while the same is true among members within the butternut group. However, only the Persian walnut seems to enjoy working both sides of the street—it will readily hybridize with members of either of the other two groups. The genetic gulf between the groups doesn't appear to be all that great, in that the various species can be grafted onto rootstock of other types. But still, this typing of the various species does function as a useful tool in understanding how the world's many walnuts differ from one another. For example, the primary two Oriental species—Manchurian walnut (*J. mandshurica*) and the Japanese walnut (*J. ailantifolia*)—appear to be of the butternut type. While their woods are all slightly denser than our native butternut, they tend to be light in color like butternut, usually with amber gold highlights. Those species found in the West Indies, primarily *J. jamaicensis*, are also of the butternut type, as is one of the species native to Central America. The two minor species found in Texas, New Mexico, and Arizona—*J. microcarpa* and *J. major*—are not important timber sources nor are they used for commercial nut production, but they yield attractively dark woods and appear to be of the black walnut type. So also are our two native West Coast species, California walnut (*J. californica*) and Hinds walnut (*J. hindsii*). In fact, most of the remaining species found from Central Mexico southward along the mountain ranges down into northwestern Argentina,



PHOTO BY KENNETH GADWAY

This bowl was made from butternut (*Juglans cinera*) harvested in the Adirondack Mountains of New York by woodturner Kenneth Gadway.

also seem to be of the black walnut type, but they are the least understood of all the species. They tend to grow at high elevations in isolated and very remote valleys of the Andes. Some of them produce woods with variegated figure, reminiscent of Persian walnut, and they offer the tantalizing prospect that they might represent additional members of this third group. The Argentine species, *J. australis*, could be another such candidate, since it often produces wood with almost black striping. Unfortunately, so far there has been little horticultural experimentation with these Latin American walnuts and, also, their woods often come to market so hopelessly commingled that it is impossible to trace the exact species.

LOOKING FORWARD

Geologically speaking, all the evidence regarding the Juglandaceae points to a botanical family that is well beyond its prime, but it would still be grossly premature to begin the eulogy. Hard pressed though our remaining reserves of the more important species of walnuts may be, they are not dangerously rare. Perhaps the healthiest of them all is our own domestic black walnut. It is estimated that the current growing stock of this species throughout the forests of eastern North America approaches 15 billion board feet. True, this is at various levels of maturity and only a small fraction of it is available for immediate harvest, but this amount of growing stock suggests a healthy and self-sustaining population.

Reserves of virtually all other species of walnut are in substantially weaker shape. While once relatively plentiful, our native butternut is experiencing a blight, and its future role as a high volume cabinet wood is uncertain. Persian walnut (*J. regia*) never has been a plentiful commercial timber, but as a managed nut cultivar both in Europe and now in the western United States, the current supply of this wood should remain stable and sustainable. Known by its Spanish common name "claro," enough of a trickle of this immigrant walnut is provided by the California commercial nut-producing industry to give the species some North American notoriety and an especially respectable reputation among many West Coast woodworkers. Also, the more recent introduction of the hardier Carpathian strain of *J. regia* by nut growers in the upper Midwest suggests that in the decades to come the availability of this wood may broaden.

As for the several Oriental species, which could perhaps function as reasonable substitutes for our now dwindling butternut, virtually none of these timbers enter into the international lumber market from sources within their native range. Fortunately, there is an expanding interest among nut growers for some of the Oriental species like heartnut and various other hybrids, so supplies of these historically Asian cabinet woods may actually become more available to American woodworkers as our domestic plantations mature.

The one remaining unknown is: what will become of the Latin American walnuts? The availability of these timbers on the international market has increased in recent years, but it is a false bounty generated more by the desperate need of the source countries for hard currency exports. Walnut must struggle mightily to survive in the tropics. It manages to do so, but only in high elevation micro-environments. These Latin American species are not really plentiful anywhere, nor does there appear to be any significant effort underway to preserve their diversity.

Obviously, the picture painted by these trends is one of a tightening and ever more managed supply of virtually all of the walnut species. In the case of black walnut, it is a virtual certainty that there will be a continuing need to harvest increasingly less mature logs, with wide sapwood that must be steamed to make as much of the stock useable as possible. Also, we can expect that commercial nut growers, coaxed by the profit motive, will be placing more emphasis on the value of their trees as an eventual source of lumber. This has long been the practice with black walnut, but it will expand to other species when these "strange exotics" gain more notoriety as the valuable cabinet woods they really are. However, the supply will probably never be great, because the grower's first priority is nut production. Logging rotations will remain based on what optimizes the nut crop, rather than on cycles that might be more efficient in maximizing lumber volume.

And for we the woodworkers, it appears that what is to come will be a bit of a mixed bag. Will walnut remain available? Sure. In fact, there is every reason to believe that as the decades role by it will be coming to lumber outlets near you in an ever expanding array of exotic new flavors and colors…but at a price that will convince you to use it sparingly.

Jon Arno is a wood technologist who has written numerous articles on the topic; he lives in Troy, Michigan.